

Chapter Five

Physical Design Using SQL Server

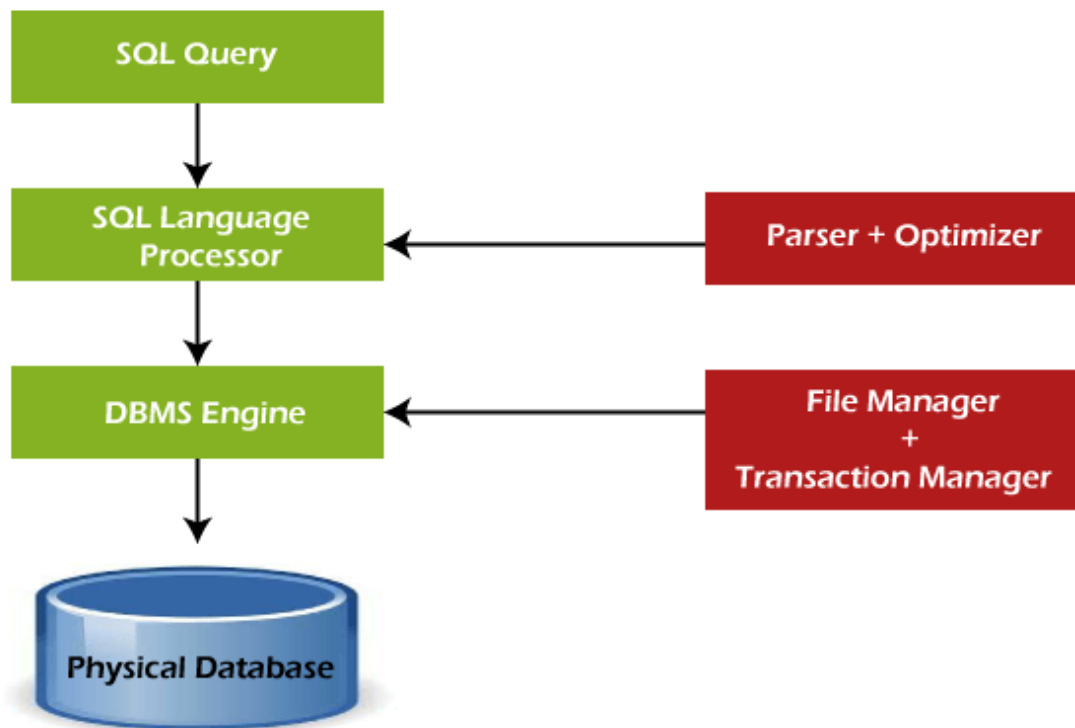
CHAPTER FIVE

PHYSICAL DESIGN USING SQL SERVER

The SQL Environment

SQL (Structured Query Language) is used to perform operations on the records stored in the database, such as updating records, inserting records, deleting records, creating and modifying database tables, views, etc. SQL is not a database system, but it is a query language. Suppose you want to perform the queries of SQL language on the stored data in the database. You are required to install any database management system in your systems, for example, Oracle, MySQL, MongoDB, PostgreSQL, SQL Server, DB2, etc.

The SQL architecture is as in figure below.



SQL Syntax

When you want to do some operations on the data in the database, then you must have to write the query in the predefined syntax of SQL. The syntax of the structured query language is a unique set of rules and guidelines, which is not **case-sensitive**. Its Syntax is defined and maintained by the ISO and ANSI standards.

Following are some most important points about the SQL syntax which are to remember:

- ❖ You can write the keywords of SQL in both uppercase and lowercase, but writing the SQL keywords in uppercase improves the readability of the SQL query.
- ❖ SQL statements or syntax are dependent on text lines. We can place a single SQL statement on one or multiple text lines.
- ❖ You can perform most of the action in a database with SQL statements.
- ❖ SQL syntax depends on relational algebra and tuple relational calculus.

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SQL Statement

SQL statements tell the database what operation you want to perform on the structured data and what information you would like to access from the database. The statements of SQL are very simple and easy to use and understand. They are like plain English but with a particular syntax. Each SQL statement begins with any of the SQL keywords and ends with the semicolon (;). The semicolon is used in the SQL for separating the multiple Sql statements which are going to execute in the same call. In this SQL tutorial, we will use the semicolon (;) at the end of each SQL query or statement.

Most Important SQL Commands and Statements

- ❖ Select Statement
- ❖ Update Statement
- ❖ Delete Statement
- ❖ Create Table Statement
- ❖ Alter Table Statement
- ❖ Drop Table Statement
- ❖ Create Database Statement
- ❖ Drop Database Statement
- ❖ Insert Into Statement
- ❖ Truncate Table Statement
 - ☐ Describe Statement
 - ☐ Distinct Clause
 - ☐ Commit Statement
 - ☐ Rollback Statement
 - ☐ Create Index Statement
 - ☐ Drop Index Statement
 - ☐ Use Statement

SQL Data Types

Data types are used to represent the nature/format of the data that can be stored in the database table. For example, in a particular column of a table, if we want to store a string type of data then we will have to declare a string data type of this column.

Data types mainly classified into three categories for every database.

- ❖ String Data types
- ❖ Numeric Data types
- ❖ Date and time Data types

Data Types in MySQL, SQL Server and Oracle Databases

CHAR(Size)	It is used to specify a fixed length string that can contain numbers, letters, and special characters. Its size can be 0 to 255 characters. Default is 1.
VARCHAR(Size)	It is used to specify a variable length string that can contain numbers, letters, and special characters. Its size can be from 0 to 65535 characters.
BINARY(Size)	It is equal to CHAR() but stores binary byte strings. Its size parameter specifies the column length in the bytes. Default is 1.
VARBINARY(Size)	It is equal to VARCHAR() but stores binary byte strings. Its size parameter specifies the maximum column length in bytes.
TEXT(Size)	It holds a string that can contain a maximum length of 255 characters.
TINYTEXT	It holds a string with a maximum length of 255 characters.
MEDIUMTEXT	It holds a string with a maximum length of 16,777,215.
LONGTEXT	It holds a string with a maximum length of 4,294,967,295 characters.

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ENUM(val1, val2, val3,...)	It is used when a string object having only one value, chosen from a list of possible values. It contains 65535 values in an ENUM list. If you insert a value that is not in the list, a blank value will be inserted.
SET(val1, val2, val3,...)	It is used to specify a string that can have 0 or more values, chosen from a list of possible values. You can list up to 64 values at one time in a SET list.
BLOB(size)	It is used for BLOBs (Binary Large Objects). It can hold up to 65,535 bytes.

MySQL Numeric Data Types

BIT(Size)	It is used for a bit-value type. The number of bits per value is specified in size. Its size can be 1 to 64. The default value is 1.
INT(size)	It is used for the integer value. Its signed range varies from -2147483648 to 2147483647 and unsigned range varies from 0 to 4294967295. The size parameter specifies the max display width that is 255.
INTEGER(size)	It is equal to INT(size).
FLOAT(size, d)	It is used to specify a floating point number. Its size parameter specifies the total number of digits. The number of digits after the decimal point is specified by d parameter.
FLOAT(p)	It is used to specify a floating point number. MySQL used p parameter to determine whether to use FLOAT or DOUBLE. If p is between 0 to 24, the data type becomes FLOAT (). If p is from 25 to 53, the data type becomes DOUBLE().
DOUBLE(size, d)	It is a normal size floating point number. Its size parameter specifies the total number of digits. The number of digits after the decimal is specified by d parameter.
DECIMAL(size, d)	It is used to specify a fixed point number. Its size parameter specifies the total number of digits. The number of digits after the decimal parameter is specified by d parameter. The maximum value for the size is 65, and the default value is 10. The maximum value for d is 30, and the default value is 0.
DEC(size, d)	It is equal to DECIMAL(size, d).
BOOL	It is used to specify Boolean values true and false. Zero is considered as false, and nonzero values are considered as true.

MySQL Date and Time Data Types

DATE	It is used to specify date format YYYY-MM-DD. Its supported range is from '1000-01-01' to '9999-12-31'.
DATETIME(fsp)	It is used to specify date and time combination. Its format is YYYY-MM-DD hh:mm:ss. Its supported range is from '1000-01-01 00:00:00' to 9999-12-31 23:59:59.
TIMESTAMP(fsp)	It is used to specify the timestamp. Its value is stored as the number of

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	seconds since the Unix epoch('1970-01-01 00:00:00' UTC). Its format is YYYY-MM-DD hh:mm:ss. Its supported range is from '1970-01-01 00:00:01' UTC to '2038-01-09 03:14:07' UTC.
TIME(fsp)	It is used to specify the time format. Its format is hh:mm:ss. Its supported range is from '-838:59:59' to '838:59:59'
YEAR	It is used to specify a year in four-digit format. Values allowed in four digit format from 1901 to 2155, and 0000.

SQL Server Data Types

SQL Server String Data Type

char(n)	It is a fixed width character string data type. Its size can be up to 8000 characters.
varchar(n)	It is a variable width character string data type. Its size can be up to 8000 characters.
varchar(max)	It is a variable width character string data types. Its size can be up to 1,073,741,824 characters.
text	It is a variable width character string data type. Its size can be up to 2GB of text data.
nchar	It is a fixed width Unicode string data type. Its size can be up to 4000 characters.
nvarchar	It is a variable width Unicode string data type. Its size can be up to 4000 characters.
ntext	It is a variable width Unicode string data type. Its size c

Query Language

The SQL query language has set queries statements/commands specified in the above section. These queries can be classified into two based on their operations. The classification is as

1. **DDL (data definition language) and**
2. **DML(data manipulation language)**

1. DDL (data definition language)

These set of query are used to define the database structure /schema. The DDL query statement include

- ❖ **Create Statement**
- ❖ **Alter Statement**
- ❖ **Drop Statement**

SQL CREATE query/command statement

The SQL **create command/query** is used to define the database objects. The basic database objects are:-

- ❖ The **database** it self
- ❖ The database table
- ❖ Views which is a virtual table
- ❖ Stored procedures
- ❖ Triggers

Creating Database Objects

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The organization of a database involves many different objects. All objects of a database can either be physical or logical. The physical objects are related to the organization of the data on the physical device (disk). Database Engine's physical objects are files and file groups. Logical objects represent a user's view of a database. **Databases, tables, columns, and views (virtual tables) are examples of logical objects.** The first database object that has to be created is a database itself. Database Engine manages both system and user databases. An authorized user can create user databases, while system databases are generated during the installation of the database system. System databases are

- ❖ **master**
- ❖ **tempdb**
- ❖ **model**
- ❖ **msdb**
- ❖ **resource**

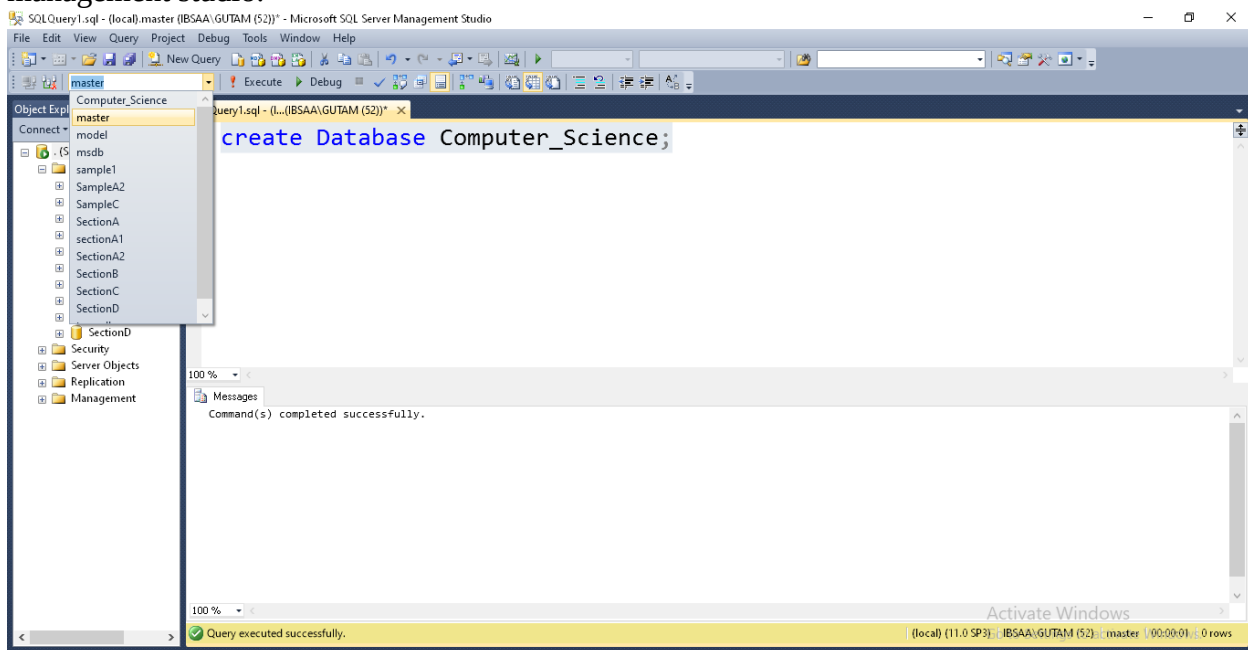
This section describes the creation, alteration, and removal of user databases.

Creation of a Database

Two basic methods are used to create a database. The first method involves using Object Explorer in SQL Server Management Studio. The second method involves use of the command statement **CREATE DATABASE**. This statement has the following general form. The SQL CREATE DATABASE statement is used by a developer to create a database. Let's see the syntax of SQL CREATE DATABASE:

CREATE DATABASE database_name;

The implementation of the above command is shown in the following figure on SQL server 2012 management studio.



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SQL DROP Database

SQL DROP statement is used to delete or remove indexes from a table in the database. If we want to delete or drop an existing database in a SQL schema, we can use SQL DROP DATABASE. *Let's see the syntax of SQL DROP DATABASE:*

DROP DATABASE database_name;

***Example:- drop database Sample;
drop table Student;***

SQL RENAME Database

In some situations, database users and administrators want to change the name of the database for some technical reasons. So, the **Rename Database** statement in SQL is used to change the name of the existing database. Sometimes, the Rename Database statement is used because the developers think that the original name is not more relevant to the data of the database, or they want to give a temporary name to that database.

Syntax of Rename Database in SQL

ALTER DATABASE old_database_name MODIFY NAME = new_database_name;

Suppose we want to rename the Student Database. For this, we have to type the following query in SQL:

ALTER DATABASE Student MODIFY NAME = College;

This query will change the name of the database from Student to College. To run this query, we must ensure that the database Student exists in the current database server. If not, then it will show an error in the output.

Creation of SQL Table

Table is a collection of data, organized in terms of rows and columns. In DBMS term, table is known as relation and row as tuple. Table is the simple form of data storage and it is a database logical object. A table is also considered as a convenient representation of relations. Table variables are used to store a set of records. So declaration syntax generally looks like CREATE TABLE syntax.

create table tablename ('column names' data type);

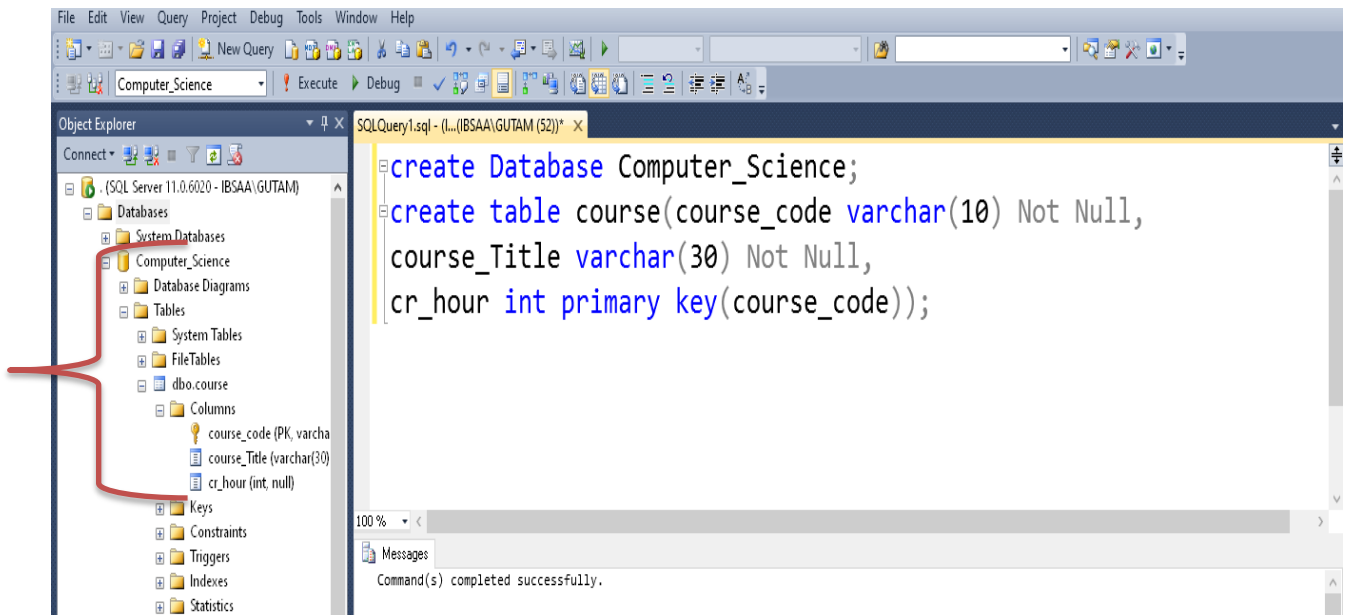
example:- create table course(course_code varchar(10) Not Null, course_Title varchar(30) Not Null, cr_hour int primary key(course_code));

Note:- In the create table command in the above example,

- ❖ The command defines the physical/Internal schema of the database table.
- ❖ The command includes constraints like primary key and disallowing null values
- ❖ Integrity is defined using the data type and length
- ❖ Further the database in which this table is created should be specified as in figure below.

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The database table **course** is created in the **computer_Science** as shown in the figure. The data type of the columns may vary from one database to another. For example, NUMBER is supported in Oracle database for integer value whereas INT is supported in MySQL.

SQL DROP TABLE

A SQL DROP TABLE statement is used to delete a table definition and all data from a table. This is very important to know that once a table is deleted all the information available in the table is lost forever, so we have to be very careful when using this command. Let's see the syntax to drop the table from the database.

```
DROP TABLE "table_name";
```

2. DML(data manipulation language)

These set of query are used to manipulate/change the data content of database structure /schema instances. Some of the DML query statements are:-

- ❖ **Select Statement**
- ❖ **Update Statement**
- ❖ **Delete Statement**
- ❖ **Insert Into Statement**

SQL Insert Statement

SQL INSERT statement is a SQL query. It is used to insert a single or a multiple records in a table.

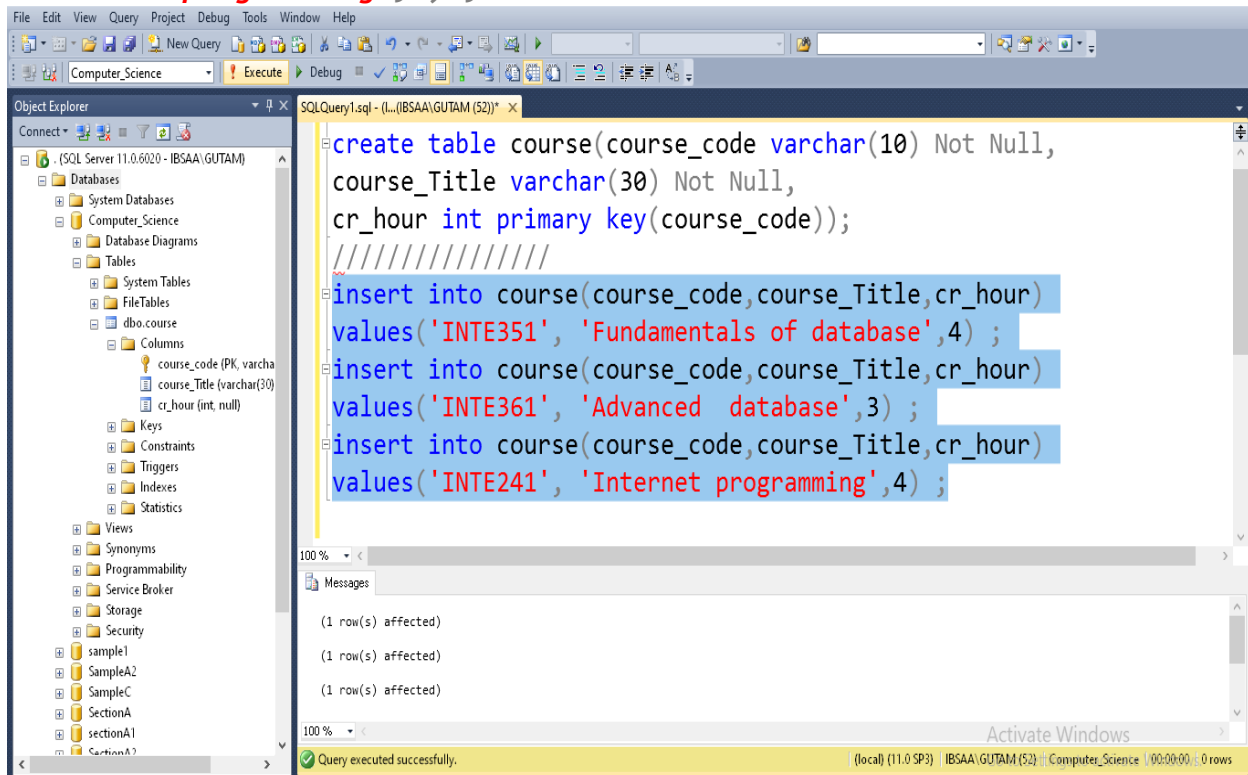
```
INSERT INTO table_name (column1, column2, column3....)  
VALUES (value1, value2, value3.....);
```

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Example:- this example shows how to insert multiple rows.

```
insert into course(course_code,course_Title,cr_hour)values('INTE351',  
'Fundamentals of database',4) ;  
insert into course(course_code,course_Title,cr_hour)values('INTE361',  
'Advanced database',3) ;  
insert into course(course_code,course_Title,cr_hour)values('INTE241',  
'Internet programming',4) ;
```



Note:- the above shows how to write *the Insert into* command and how to execute.

SQL SELECT command/query

The most commonly used SQL command is **SELECT statement/query**. It is used to query the database and retrieve selected data that follow the conditions we want. In simple words, we can say that the select statement used to query or retrieve data from a table in the database. The select statement can be written as in the following expression.

```
SELECT expression/s  
FROM table/s  
WHERE conditions;
```

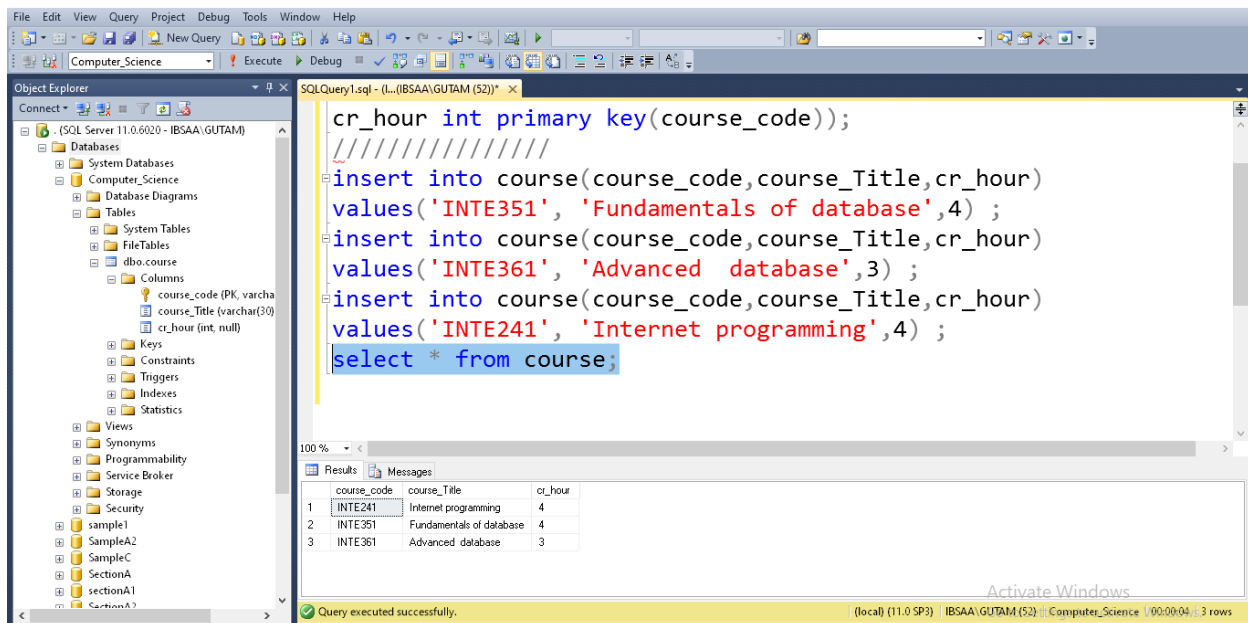
Example:- the following example shows the different ways of using the select command.

```
select * from StudentD;  
select * from course;  
select * from Department;  
select * from course where cr_hour=3;
```

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```
select * from course where cr_hour=4;
Select FName,LName,course_Title from StudentD,course;
Select FName,LName from StudentD
Select FName,LName,course_Title from StudentD,course Where
StudentD.course_code=course.course_code;
select COUNT(FName) from StudentD;
```



Note:-

- ❖ (*) expression in the select command means all contents(rows and columns) from a specified tables.
 - `select * from course;`
- ❖ It is possible to filter out columns from a given table.
 - `Select FName,LName from StudentD`
- ❖ It is possible to filter out rows from a given table.
 - `select * from course where cr_hour=4;`
- ❖ It is possible to display values from two or more tables.
 - `Select FName,LName,course_Title from StudentD,course Where StudentD.course_code=course.course_code;`
- ❖ Select command can be used with functions like `COUNT()`, `Top()`
 - `select COUNT(FName) from StudentD;`
- ❖ Select command can be used with `JOIN`
 - `Select FName,LName,course_Title from StudentD Join course ON StudentD.course_code=course.course_code;`

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SQL SELECT COUNT

The **SQL COUNT()** function is used to return the number of rows in a query. The **COUNT()** function is used with SQL **SELECT** statement and it is very useful to count the number of rows in a table having enormous data.

For example: If you have a record of the voters in selected area and want to count the number of voters then it is very difficult to do it manually but you can do it easily by using the SQL **SELECT COUNT** query.

```
SELECT COUNT (expression)
FROM tables
WHERE conditions;
```

SQL SELECT TOP

The SQL **SELECT TOP** Statement is used to select top data from a table. The top clause specifies that how many rows are returned.

SQL WHERE

The **WHERE** clause in SQL select statement is used as a mechanism of defining condition/s in data manipulation language(DML).WHERE clause is not mandatory clause of SQL DML statements. But it can be used to limit the number of rows affected by a SQL DML statement or returned by a query. Actually, it filters the records. It returns only those queries which fulfill the specific conditions.

WHERE clause is used in SELECT, UPDATE, DELETE statement etc.

```
SELECT column1, column 2, ... column n
FROM table_name
WHERE [conditions]
```

WHERE clause uses some conditional selection

=	equal
>	greater than
<	less than
>=	greater than or equal
<=	less than or equal
< >	not equal to

SQL DELETE

The **SQL DELETE statement** is used to delete rows from a table. Generally **DELETE** statement removes one or more records from a table.

SQL DELETE Syntax

Let's see the Syntax for the SQL **DELETE** statement:

```
DELETE FROM table_name [WHERE condition];
```

Example: **Delete from course where course_code= 'INTE351' ;**

Here **table_name** is the table which has to be deleted. The **WHERE** clause in SQL **DELETE** statement is optional here.

Note:- the delete command is applied to a specific row.

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SQL UPDATE

The SQL commands (*UPDATE* and *DELETE*) are used to modify the data that is already in the database. The SQL *DELETE* command uses a *WHERE* clause.

SQL UPDATE statement is used to change the data of the records held by tables. Which rows is to be updated, it is decided by a condition. To specify condition, we use **WHERE** clause.

The *UPDATE* statement can be written in following form:

UPDATE table_name SET [column_name1= value1,... column_nameN = valueN] [WHERE condition.

*UPDATE table_name
SET column_name = expression
WHERE conditions*

Example:

Update course set cr_hour=4 where course_code='CoSc311';